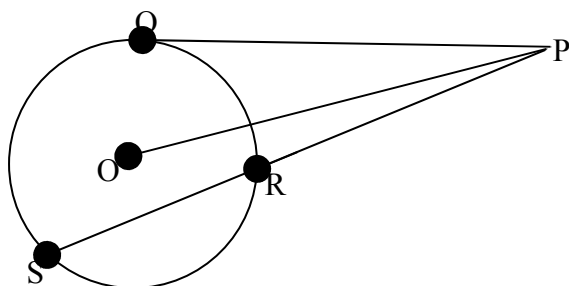


Question 1

- a) For the parabola $y = (x+2)^2 - 5$, state the vertex and axis of symmetry. 2
- b) The line l has an x-intercept of -6 and y-intercept of 5 . 3
 The line, k has an x-intercept of 3 and y-intercept of 4 .
 Find the acute angle between the lines giving your answer to nearest minute.
- c) (i) On the same axes, sketch the curves of $y = x^2$ and $y = |x|$ 2
 (ii) Hence, or otherwise, solve $x^2 < |x|$ 1

d)



O is the centre of a circle of radius 6cm, PQ is a tangent and PRS is a secant. $PO = 10\text{cm}$, $PR = 6\text{cm}$. Find SR.

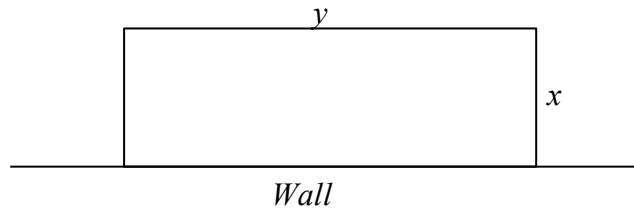
Question 2

- a) The roots of the quadratic equation $x^2 + 4x + 2 = 0$ are α and β .
 (i) Find $(\alpha - 2)(\beta - 2)$. 2
 (ii) Find $\alpha^2 + \beta^2$ 2
 (iii) Hence, show that the value of $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = -5$ 2
- b) If A is acute, B is obtuse and $\cos A = \frac{5}{13}$ and $\sin B = \frac{3}{5}$, 2
 find $\operatorname{cosec} A \tan B$.
- c) Find the equation of the tangent to the curve $y = 2x\sqrt{x+1}$ 4
 at the point where $x = 3$.

Question 3

- a) A is the point $(5, 0)$ and O is the origin. Given that the point 4
 $B(x, y)$ lies on the line $y = 1 - 3x$ and that OB is perpendicular to AB,
 find the coordinates of B.

- b) Farmer Joe decided to make a rectangular chicken run using an existing wall as one side.



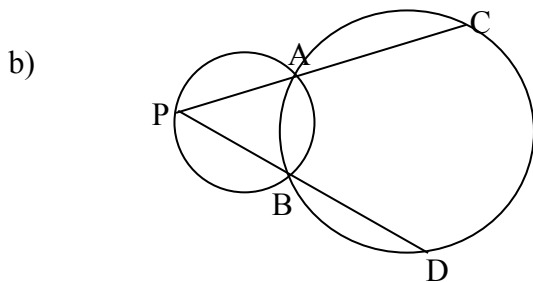
He has 18m of wire netting to use.

- (i) Show that $y = 18 - 2x$ 1
- (ii) Show that the area, $A \text{ m}^2$, of the run is given by 1
- $$A = 18x - 2x^2$$
- (iii) Find the maximum possible area of the run. 3
- (Use quadratic theory only)

- c) For what values of k , does the quadratic equation $kx^2 + 4x + k + 3 = 0$, have real roots?

Question 4

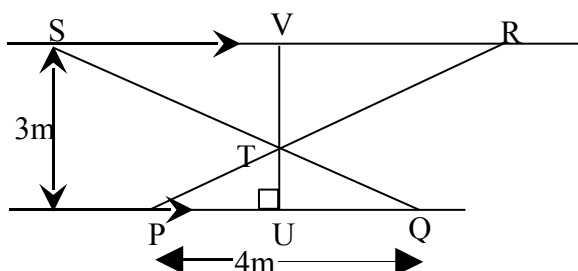
- a) $y = x^2 + bx + c$
- (i) Find $\frac{dy}{dx}$ 1
- (ii) The line $x + 7y - 15 = 0$ is a normal to the curve $y = x^2 + bx + c$ at the point $(1, 2)$. Find b and c . 4



Copy this diagram on to your answer sheet. In the diagram PAC and PBC are straight lines. Prove that DC is parallel to the tangent P.

- c) Solve $2\cos^2 2\theta - \cos 2\theta - 1 = 0$ for $0 \leq \theta \leq 360$.

Question 5



In the diagram, PQ and SR are parallel railings which are 3 Metres apart. The points P and Q are fixed 4m apart on the Lower railing. Two crossbars PR and QS intersect at T as shown in the diagram. The line perpendicular to PQ intersects PQ at U and SR at V. the length of UT is y metres.

(i) Show that ΔSRT is similar to ΔQPT . 3

(ii) Hence explain why $\frac{SR}{PQ} = \frac{VT}{UT}$ 1

(iii) Show that $SR = \frac{12}{y} - 4$ 1

(iv) Hence, show that the total area, A, of ΔQPT 2

and ΔRTS is $A = 4y - 12 + \frac{18}{y}$

b) If $a^x = 7$ and $a^{x+3} = 56$, find the values of a^{x-3} 2

c) A and B have coordinates $(-1, 7)$ and $(5, -2)$ respectively, P divides the interval AB in the ratio $k: 1$

(i) Write down the coordinates of P in terms of k . 1

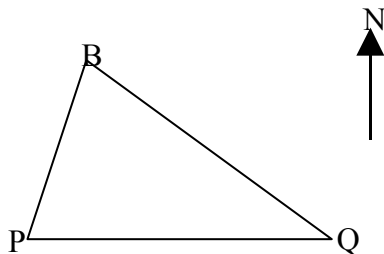
(ii) Hence find AP : PB when P lies on the line $5x - 4y - 1 = 0$. 2

Question 6

a)(i) Determine whether $f(x) = x^2 - 2x + 2$ is positive definite, negative definite or neither. 2

(ii) Hence, solve $\frac{x^2 - 2x + 2}{3 - x}$ 2

b) Two observers P and Q are 150m apart. The observer at P finds the bearing of Q to be $087^\circ T$ and a distant tower AB, to have a bearing of $006^\circ T$. The observer at Q finds the bearing of the tower to be $357^\circ T$.

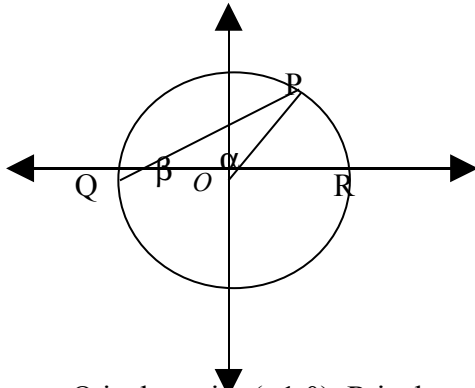


(i) Copy the diagram onto your answer sheet and add any given information. 1

(ii) Show that $BQ = \frac{150 \sin 81^\circ}{\sin 9^\circ}$ 2

- (iii) The observer at Q finds the angle of elevation of the top of the tower to be 7° . Find the height of the tower correct to the nearest metre. 2
- c) The line $3x + 4y + 7 = 0$ is a tangent to a circle with centre $(2,1)$. Find the equation of the circle. 3

Question 7



In the diagram, Q is the point $(-1,0)$, R is the point $(1,0)$, and P is another point on the circle with centre O and radius 1 unit. Let $\angle POR = \alpha$ and $\angle PQR = \beta$, and let $\tan \beta = m$.

- (i) Explain why $\triangle OPQ$ is isosceles, and hence deduce that $\alpha = \beta$ 2
- (ii) Find the equation of the line PQ. 1
- (iii) Show that the coordinates of P and Q are solutions of the equation $(1+m^2)x^2 + 2m^2x + m^2 - 1 = 0$. 2
- (iv) Using this equation, find the coordinates of P in terms of m . 3

- (v) Hence deduce that 2

$$\tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

- (vi) Hence, using part(v) only, show that 2
- $$\tan 15^\circ = 2 - \sqrt{3}$$

Question 1

a) $y = (x+2)^2 - 5$

Vertex = $(-2, -5)$

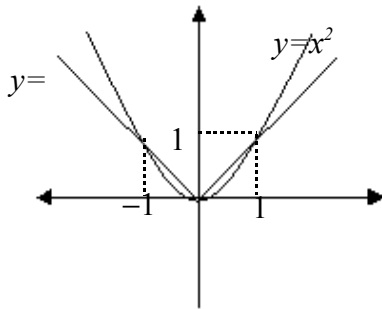
Axis of symmetry $x = -2$

b) $m_l = \frac{5}{6}$ $m_k = \frac{-4}{3}$

$$\tan \theta = \left| \frac{m_k - m_l}{1 + m_k m_l} \right| = \left| \frac{\frac{-4}{3} - \frac{5}{6}}{1 + \frac{-4}{3} \times \frac{5}{6}} \right| = 19 \frac{1}{2}$$

$\angle \theta = 87^\circ 4'$

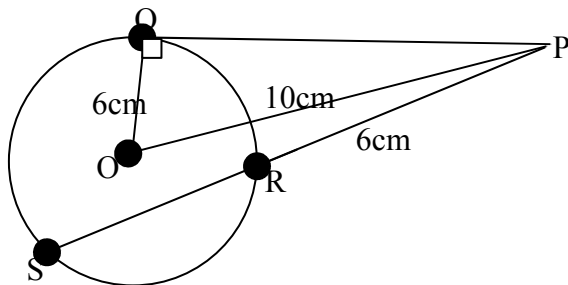
c)(i)



(ii) For $x^2 < |x|$,

$-1 < x < 0$ or $0 < x < 1$

d)



$\angle POQ = 90^\circ$ (tangent is perpendicular to radius)

$10^2 = PQ^2 + 6^2$ (Pythag. Theorem)

$\therefore PQ = 8\text{cm}$

$PQ^2 = SP \times PR$ (Square of tangent = product of intercept of secants)

$8^2 = (SR + 6) \times 6$

$SR = 4 \frac{2}{3}$

Question 2

a) $x^2 + 4x + 2 = 0$

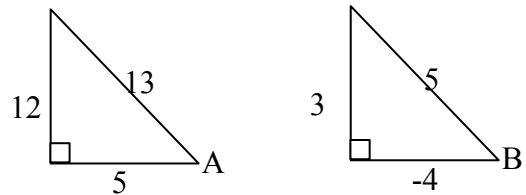
$\alpha + \beta = -4$ and $\alpha\beta = 2$

(1) $(\alpha - 2)(\beta - 2) = \alpha\beta - 2(\alpha + \beta) + 4$
 $= 2 - 2 \times -4 + 4 = 14$

(ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= (-4)^2 - 2 \times 2 = 12$

(iii) $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{\alpha^3 + \beta^3}{\alpha^3 \beta^3}$
 $= \frac{(\alpha + \beta)^3 - (\alpha^2 + \beta^2 - \alpha\beta)}{\alpha^3 \beta^3}$
 $= \frac{(-4)^3 - (12 - 2)}{2^3} = -5$

b)



$\text{Cosec A} \tan B = \frac{13}{12} \times \frac{3}{-4} = -\frac{13}{16}$

c) $y = 2x\sqrt{x+1}$

$\frac{dy}{dx} = 2x(x+1)^{\frac{1}{2}}$
 $= 2(x+1)^{\frac{1}{2}} + 2x \times \frac{1}{2}(x+1)^{-\frac{1}{2}}$
 $= 2\sqrt{x+1} + \frac{x}{\sqrt{x+1}}$

coordinates of tangents at $x = 3$

$m_t = 2\sqrt{4} + \frac{3}{\sqrt{3+1}} = \frac{11}{2}$

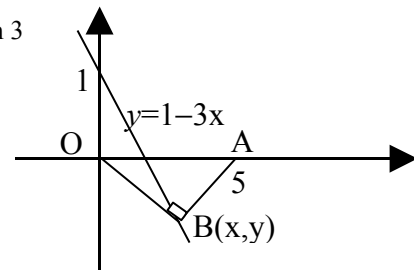
when $x = 3$, $y = 2 \times 3 + \sqrt{4} = 12$

Equation of tangent

$y - 12 = \frac{11}{2}(x - 3)$

$11x - 2y - 9 = 0$

Question 3



$m_{OB} = \frac{y}{x}$ $m_{AB} = \frac{y - 0}{x - 5}$

as $OB \perp AB$

$\frac{y}{x} \times \frac{y}{x - 5} = -1$

$$y^2 = -x^2 + 5x \dots\dots\dots \textcircled{1}$$

But B(x, y) lies on $y = 1 - 3x$
 $\therefore (1 - 3x)^2 = -x^2 + 5x$

$$1 - 6x + 9x^2 = -x^2 + 5x$$

$$10x^2 - 11x + 1 = 0$$

$$(10x - 1)(x - 1) = 0$$

$$x = \frac{1}{10} \text{ or } 1$$

$$\therefore B\left(\frac{1}{10}, \frac{7}{10}\right) \text{ or } (1, -2)$$

b)(i) perimeter = $2x + y$
 ie. $2x + y = 18$
 hence, $y = 18 - 2x$

(ii) Area A = xy
 $= x(18 - 2x)$
 $A = 18x - 2x^2$

(iii) As this is a quadratic function, $a = -2 < 0$
 concave down.

Hence, the maximum value will exist.

$$\text{Axis of symmetry } x = \frac{-b}{2a} = \frac{-(-18)}{2(-2)} = \frac{9}{2}$$

$$x = \frac{9}{2}$$

$$\therefore A = \frac{9}{2} \left(18 - 2 \times \frac{9}{2}\right) = 40\frac{1}{2}$$

$$\therefore \text{Maximum area} = 40\frac{1}{2} \text{ m}^2$$

c) $kx^2 + 4x + k + 3 = 0$
 For real roots $\Delta \geq 0$
 ie. $4^2 - 4 \times k(k+3) \geq 0$
 $16 - 4k^2 - 12k \leq 0$
 $4k^2 + 12k - 16 \geq 0$
 $k^2 + 3k - 4 \geq 0$
 $(k+4)(k-1) \geq 0$

$$-4 \leq x \leq 1$$

Question 4

a)(i) $\frac{dy}{dx} = 2x + b$

(ii) $x + 7y - 15 = 0$
 $y = -\frac{1}{7}x + \frac{15}{7}$

$$m_{\text{line}} = -\frac{1}{7} = m_{\text{normal}}$$

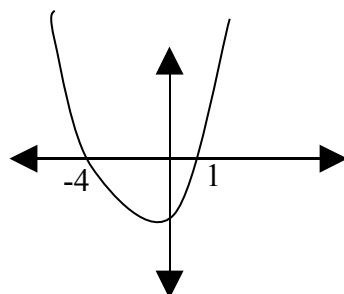
$$m_{\text{tangent}} = 7$$

Gradient of tangent at (1,2) = $2 \times 1 + b = 7$
 $\therefore b = 5$

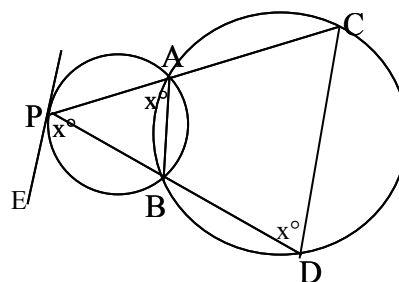
(1,2) is on the curve.

$$\text{So, } 2 = 1^2 + 5 \times 1 + c$$

$$\therefore c = -4$$



b)



Let $\angle EPB = x^\circ$

- $\angle EPB = \angle PAB = x^\circ$ ($\angle$ bet. Tangent and chord = $\angle$ in alternate segment)
 - $\angle PAB = \angle BDC = x^\circ$ (Ext. $\angle$ of a cyclic quadrilateral = opp. Interior angle)
- $\therefore \angle EPB = \angle BDC = x^\circ$ (Alternate angles)
 $\therefore EP \parallel CD$
 ie. tangent at P is parallel to CD.

c) $2\cos^2 2\theta - \cos 2\theta - 1 = 0$ for $0 \leq \theta \leq 360$.

$$2\cos^2 2\theta - \cos 2\theta - 1 = 0$$

$$(2\cos 2\theta + 1)(\cos 2\theta - 1) = 0$$

$$\therefore \cos 2\theta = -\frac{1}{2} \text{ or } \cos 2\theta = 1$$

$$\angle 2\theta = 120^\circ, 240^\circ, 480^\circ, 600^\circ \text{ or}$$

$$0^\circ, 360^\circ, 720^\circ$$

$$\therefore \theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ \text{ or } 0^\circ, 180^\circ, 360^\circ$$

Question 5

a) (i) In ΔSRT and QPt ,

- $\angle STR = \angle PTQ$ (Vert. Opp. Angles)
 - $\angle VST = \angle TQP$ (alternate angles equal as $SR \parallel PQ$)
- $\therefore \Delta SRT \parallel \Delta QPT$ (equiangular)

(ii) $\frac{SR}{PQ} = \frac{VT}{UT}$ (corresponding sides of similar triangles are proportional. $\therefore$ heights are proportional)

(iii) $\frac{SR}{4} = \frac{3 - y}{y} = \frac{3}{y} - 1$

$$\therefore SR = \frac{12}{y} - 4$$

(iv) Area = $\Delta PTQ + \Delta RTS$

$$= \frac{1}{2} \times 4 \times y + \frac{1}{2} \left(\frac{12}{y} - 4\right) (3 - y)$$

$$= 2y + \left(\frac{6}{y} - 2\right) (3 - y)$$

$$= 4y - 12 + \frac{18}{y}$$

b) $a^x = 7$ $a^{x+3} = 56$

$$\therefore \frac{a^{x+3}}{a^x} = \frac{56}{7}$$

ie. $a^3 = 8 \Rightarrow a = 2$

$$\therefore a^{x-3} = a^x \times a^{-3}$$

$$= 7 \times 8^{-1} = \frac{7}{8}$$

c)(i) $(-1, 2) \quad k : 1 \quad (5, -2)$

$$P = \left(\frac{5k-1}{k+1}, \frac{-2k+7}{k+1} \right)$$

(ii) As P lies on line $5x-4y-1=0$,

$$5 \left(\frac{5k-1}{k+1} \right) - 4 \times \left(\frac{-2k+7}{k+1} \right) - 1 = 0$$

$$25k - 5 + 8k - 28 - k - 1 = 0$$

$$32k - 34 = 0$$

$$\therefore k = \frac{34}{32} = \frac{17}{16}$$

$$\therefore AP : PB = \frac{17}{16} : 1 = 17 : 16$$

Question 6

a)(i) $f(x) = x^2 - 2x + 2$

$$\Delta = (-2)^2 - 4 \times 1 \times 2 = -4$$

$\therefore \Delta < 0$ and as $a = 1 > 0$, $f(x)$ is a positive definite.

(ii) $\frac{x^2 - 2x + 2}{3 - x} \leq 0$

as $3 - x \neq 0$ ie. $x \neq 3$

Multiplying by $(3-x)^2$ on both sides,

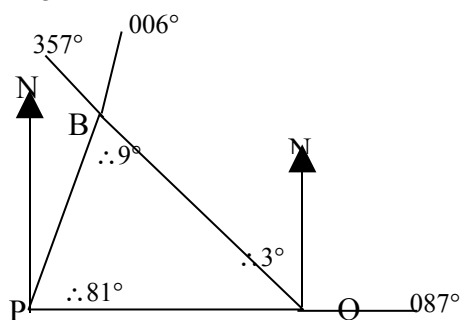
$$(3-x)(x^2 - 2x + 2) \leq 0$$

But, using part(ii) $x^2 - 2x + 2 > 0 \quad \forall x$.

$$\therefore 3-x < 0$$

$$\Rightarrow x > 3$$

b)(i)



(ii) $\angle QPB = 87^\circ - 6^\circ = 81^\circ$

$$\angle BQN = 360 - 357 = 3^\circ$$

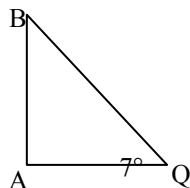
$$\angle BQP = 357 - (87 + 180) = 90^\circ$$

$$\therefore \angle PBQ = 9^\circ$$

$$\therefore \frac{BQ}{\sin 81} = \frac{150}{\sin 9}$$

$$\therefore BQ = 947.06\text{m}$$

(iii)



$$\tan 7^\circ = \frac{AB}{BQ} = \frac{947.06}{BQ}$$

$$\therefore BQ = 947.06 \times \tan 7^\circ = 116\text{m}$$

c) Perpendicular distance = $\left| \frac{3 \times 2 + 4 \times 1 + 7}{\sqrt{3^2 + 4}} \right| = \frac{17}{5}$

Centre $(2, 1)$

$$\therefore \text{Equation of the circle} = (x-2)^2 + (y-1)^2 = \left(\frac{17}{5} \right)^2$$

Question 7

a) $OP = OQ$ (equal radii)

$\therefore \triangle OPQ$ is isosceles

$\angle OPQ = \angle PQO = \beta$ (base angles equal as Δ is isosceles)

So, $\alpha = \beta + \beta$ (ext. $\angle$ = sum of opp. Int. $\angle$ s)

$\alpha = 2\beta$ (angle at the centre = twice $\angle$ at the circumference of arc PR)

b) As PQ has slope $m_{PQ} = \tan \beta$, $Q = (-1, 0)$

$\therefore$ equation is $y - 0 = m(x + 1)$

$$y = mx + m$$

c) the x-value of P and Q are the x-values of the simultaneous equations.

$$x^2 + y^2 = 1 \quad \dots\dots\dots(1)$$

$$y = mx + m \quad \dots\dots\dots(2)$$

$$\text{So, } x^2 + (mx+m)^2 = 1$$

$$x^2 + m^2x^2 + 2m^2x + m^2 = 1$$

$$\Rightarrow (1+m^2)x^2 + 2m^2x + m^2 - 1 = 0$$

d) the x-values of P is x_p .

But x-value of Q is $x_Q = -1$

$$\text{Solving (c) } x = -1 \text{ or } \frac{1-m^2}{1+m^2}$$

$$\text{Using } \alpha\beta = \frac{c}{a}, \quad -1 \times x_p = \frac{m^2 - 1}{1 + m^2}$$

$$\Rightarrow x_p = \frac{1 - m^2}{1 + m^2}$$

$$\therefore y = m \left(\frac{1 - m^2}{1 + m^2} \right) + m$$

$$= \frac{2m}{1 + m^2}$$

$$\therefore P = \left(\frac{1 - m^2}{1 + m^2}, \frac{2m}{1 + m^2} \right)$$

e) As $\alpha = 2\beta$

$$\therefore \tan \alpha = \tan 2\beta$$

$$\text{ie. } m_{OP} = \tan \alpha$$

$$\text{So, } \tan 2\beta = m_{OP}$$

$$= \frac{2m}{1 + m^2} \div \frac{1 - m^2}{1 + m^2}$$

$$= \frac{2m}{1 - m^2}$$

$$\therefore \tan 2\beta = \frac{2 \tan \beta}{1 - \tan^2 \beta}$$

f) Let $\beta = 15^\circ$

$$\text{So, } \tan 30 = \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ}$$

$$\text{ie. } \frac{1}{\sqrt{3}} = \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ}$$

$$\tan^2 15^\circ + 2\sqrt{3} \tan 15^\circ - 1 = 0$$

Solving the quadratic equation,

$$\tan 15^\circ = \pm(2 - \sqrt{3})$$

$$\text{But } \tan 15^\circ > 0, \therefore \tan 15^\circ = 2 - \sqrt{3}$$